

SOOHYUN CHOI

Integrated M.S./Ph.D. Student · Reinforcement Learning & Sequential Decision Making
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RESEARCH PROFILE

My research studies reinforcement learning and sequential decision making through iterative policy-space refinement and generative long-horizon planning. I analyze how the proximal step size τ induces distinct regimes in multi-step policy improvement, and I develop goal-conditioned state-space bridges and flows through PathBridger and its PathFlower extension.

EDUCATION

Hanyang University 2024–Present
Integrated M.S./Ph.D. Program in Electronic Engineering
Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong

MANUSCRIPT

S. Choi, S. Cho, and S. Hong, "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning."
Under Review, AAAI 2027.

RESEARCH EXPERIENCE

PathBridger — Offline Goal-Conditioned Reinforcement Learning 2026

- Developed a hierarchical offline GCRL approach that constructs explicit state-space bridges, decodes them into short executable action chunks via inverse dynamics, and replans during execution.
- Evaluated long-horizon navigation and manipulation behavior across OGBench tasks.

Multi-step Proximal Policy Improvement & PathFlower 2026–Present

- Analyzing how the proximal step size τ governs recursively re-centered policy refinement, using sweeps and mathematical diagnostics to characterize local-flow, nonlocal finite-step, and over-refinement regimes.
- Extending PathBridger toward PathFlower, which replaces an explicit subgoal with final-goal-conditioned state flows for generative long-horizon planning.

SK hynix–Hanyang University Industry–Academic Research Project 2024–2025

- Contributed to machine-learning- and transfer-function-based modeling for circuit-interconnect waveform and delay estimation, including model validation and industry technical exchanges.

HONORS & FUNDING

- Outstanding Graduation Project Award — KRW 2,000,000 award
- Excellence Award, College of Engineering Capstone Design — KRW 1,000,000 award
- BK Research Assistantship (BK-RA), Hanyang University — 2025

SELECTED GRADUATE COURSEWORK

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory

Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I